The seven faces of the collision residue: a unified programme for chiral algebras, 3d holomorphic-topological QFT, and Calabi–Yau quantum groups

Raeez Lorgat

Abstract

For every chirally Koszul chiral algebra $\mathcal{A}$ in the standard landscape, the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of the universal Maurer–Cartan element admits an *infinite family* of presentations parametrized by the Drinfeld Grothendieck–Teichmüller torsor GRT. Each associator $\Phi \in \mathrm{GRT}_1(\mathbb{Q})$ produces a face F_Φ; the bar-intrinsic face F_1 is the identity element of the torsor; Brown's motivic Drinfeld coaction supplies the motivic face F_8 (MZV-valued); Willwacher's GRT-action on the Kontsevich graph complex supplies the operadic face F_9. The seven historically-labeled faces –- twisting morphism (bar-cobar), line-operator R-twist (DNP), classical PVA r-matrix (KZ), GZ26 commuting differentials, Yangian r-matrix (Drinfeld 1985), Sklyanin bracket (STS 1983), Gaudin generator (FFR 1994) –- are seven orbit representatives under the GRT-action. as a twisting morphism in the bar-cobar framework, as an R-matrix governing the meromorphic tensor product of line operators (Dimofte–Niu–Py), as a classical Poisson vertex algebra r-matrix in the Khan–Zeng 3d sigma-model framework, as the generating function of the Gaiotto–Zeng commuting differential operators on the sphere, as the classical limit of the Yangian R-matrix (Drinfeld 1985), as the tensor generating the Sklyanin Poisson bracket (Semenov-Tian-Shansky 1983), and as the generating function of the Gaudin Hamiltonians (Feigin–Frenkel–Reshetikhin 1994). We prove that the seven listed presentations determine the same GRT-orbit of an element of $\mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$, that the torsor action is realized concretely on affine Kac–Moody as Casimir rescaling $r(z) = k\,\Omega/z \;\rightleftharpoons\; \Omega/((k+h^\vee)z)$ (the $\mathrm{GRT}^{\mathrm{fin}}$-rescaling is a tensor identity, not a chained equality; Section 4), and that the agreement of any two representatives is controlled by three independent numerical invariants $(p_{\max}, k_{\max}, r_{\max})$ that classify the standard landscape into four shadow classes G, L, C, M. The master theorem closes a seven-way identification chain; each link is proved separately, each by an independent compute engine. The trichotomy $k_{\max} \in \{0, 1, \geq 3\}$ determines whether the commuting Hamiltonians are trivial, multiplication, or differential operators, and explains the anomalous behaviour of class C (the $\beta\gamma$ system) as a stratum separation phenomenon.

1 Introduction

Seven papers, seven frameworks, one object; all projections of a single E_1 datum.

The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the degree-2 component of the E_1 Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ in the ordered convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$. Its Σ_n-coinvariant projection $\kappa(\mathcal{A})$ is recovered from $\mathrm{av}(r(z))$ in the abelian and scalar families, and from $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody: $\kappa(\mathcal{H}_k) = k$ for Heisenberg, $\kappa(\mathrm{Vir}_c) = c/2$ for Virasoro, $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ for affine Kac–Moody (the Sugawara shift $\dim(\mathfrak{g})/2$ arises from the simple-pole self-contraction channel). The collision residue $r(z)$ retains the full spectral-parameter dependence that κ discards; the seven faces are seven ways of reading this E_1 data.

The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of the universal Maurer–Cartan element of a chirally Koszul chiral algebra $\mathcal{A}$ appears under different names in seven different literatures. In the bar-cobar framework of Ginzburg–Kapranov and Hinich, it is the genus-zero binary projection of the twisting morphism $\pi_{\mathcal{A}}\colon B(\mathcal{A}) \to \mathcal{A}$. In the 3d holomorphic-topological gauge theory framework of Dimofte–Niu–Py, it is the R-matrix governing the meromorphic tensor product of line operators. In the Poisson vertex algebra sigma-model framework of Khan–Zeng, it is the classical r-matrix extracted from the λ-bracket. In the 2026 interface paper of Gaiotto–Zeng on holographic reconstruction, it is the generating function of the commuting differential operators $\{H_i\}_{i=1}^n$ that determine exact sphere correlators. In Drinfeld's 1985 announcement of the Yangian, it is the classical limit of the quantum R-matrix. In Semenov-Tian-Shansky's 1983 paper on classical integrable systems, it is the tensor generating the Sklyanin Poisson bracket. In the 1994 paper of Feigin–Frenkel–Reshetikhin on the Gaudin model, it is the generating function of the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$.

These seven presentations look different because they are written in different categorical languages: twisting morphisms, R-twisted tensor products, λ-brackets, differential operators, classical r-matrices, Poisson tensors, Gaudin Hamiltonians. The main result of this paper is that they pick out the same underlying element of $\mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$, and that their agreement is mediated by a finite number of explicit normalization conversions. The chain of identifications is F1 $\Leftrightarrow$ F2 $\Leftrightarrow$ F3 $\Leftrightarrow$ F4 $\Leftrightarrow$ F5 $\Leftrightarrow$ F6 $\Leftrightarrow$ F7; each link is proved separately, each by an independent mathematical argument, each by an independent compute engine.

The seven-face programme is not just an enumeration of equivalences. It is a classification: the shadow obstruction tower of $\mathcal{A}$ partitions the standard landscape into four classes G, L, C, M, and the seven-face identification behaves differently in each class. For classes G (Gaussian: Heisenberg, lattice VOAs) and L (Lie/tree: affine Kac–Moody), the commuting Hamiltonians are multiplication operators of order 0; for class M (mixed: Virasoro, $\mathcal{W}_N$), they are genuine differential operators of order $2N-2$; for class C (contact: $\beta\gamma$), they vanish identically. The trichotomy of operator orders $\{0, 0, \geq 2\}$ for classes $\{G \cup L, C, M\}$ exhibits class C as exceptional in a way that is not visible from the shadow depth alone. Shadow depth $r_{\max}$ is not the right invariant to predict operator order: one must use the collision depth $k_{\max} = p_{\max} - 1$, where $p_{\max}$ is the maximal OPE pole order among generating fields. For $\beta\gamma$, $p_{\max} = 1$, so $k_{\max} = 0$, so the Hamiltonians vanish.

2 The three invariants

Three distinct numerical invariants of a chiral algebra $\mathcal{A}$ measure three different structural features and need not coincide. Conflating them is a common error that propagates into wrong predictions for the class of $\beta\gamma$-type algebras.

Definition 2.1 (Generator OPE pole order)**.** Let $\mathcal{A}$ be a chiral algebra with a chosen set of generating fields $\{a_\alpha\}$. The *generator OPE pole order* of $\mathcal{A}$ is

$$p_{\max}(\mathcal{A}) \;:=\; \max_{\alpha,\beta}\Big\{ n \geq 0 \;\Big|\; a_\alpha(z)a_\beta(w) \text{ has a nonzero } (z-w)^{-n} \text{ term}\Big\}.$$

Equivalently, $p_{\max}(\mathcal{A})$ is the maximal n such that some OPE coefficient $a_{\alpha,(n-1)}a_\beta$ among generating-field pairs is nonzero.

Definition 2.2 (Collision depth)**.** The *collision depth* of $\mathcal{A}$ is

$$k_{\max}(\mathcal{A}) \;:=\; \max\Big\{ k \geq 0 \;\Big|\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \text{ has a nonzero } z^{-k} \text{ term}\Big\}.$$

Definition 2.3 (Shadow depth)**.** The *shadow depth* of $\mathcal{A}$ is

$$r_{\max}(\mathcal{A}) \;:=\; \max\big\{ r \geq 2 \;\big|\; \Theta^{\leq r}_{\mathcal{A}} \text{ has a nonzero } r\text{-degree projection}\big\} \;\in\; \{2,3,4,\infty\}.$$

Proposition 2.4 (Relations and independence)**.** *For every chiral algebra $\mathcal{A}$ in the standard landscape:*

(i) $k_{\max}(\mathcal{A}) = p_{\max}(\mathcal{A}) - 1$.

(ii) $r_{\max}(\mathcal{A})$ *is independent of* $p_{\max}(\mathcal{A})$*: the pair* $(p_{\max}, r_{\max})$ *achieves values* $(1,4)$ *for* $\beta\gamma$*,* $(2,2)$ *for Heisenberg,* $(2,3)$ *for affine* $\mathfrak{sl}_N$*,* $(4,\infty)$ *for Virasoro, and* $(2N,\infty)$ *for* $\mathcal{W}_N$*.*

(iii) *The pair* $(r_{\max}, \text{class})$ *classifies the standard landscape into four shadow-depth classes* G*,* L*,* C*,* M*.*

Proof. Part (i) is the dlog absorption of the bar propagator: the bar differential extracts residues against $d\log(z-w) = dz/(z-w)$, so a pole of order n in the OPE produces a residue of order $n-1$ in the collision expansion.

Part (ii) is verified by explicit computation for each standard family. The $\beta\gamma$ case is the critical witness: its generator OPE $\beta(z)\gamma(w) \sim (z-w)^{-1}$ is simple-pole only, so $p_{\max}(\beta\gamma) = 1$ and $k_{\max}(\beta\gamma) = 0$; yet the composite $:\beta\gamma:$ at weight 1 has a self-OPE contributing at degree 4 to the shadow obstruction tower, so $r_{\max}(\beta\gamma) = 4$.

Part (iii) is the shadow-depth classification. Class G is characterized by termination at degree 2 (all higher shadows vanish); class L by termination at degree 3 (the Jacobi obstruction is the last nonvanishing term); class C by termination at degree 4 (the quartic contact class is the last nonvanishing term); class M by non-termination (all degrees carry nontrivial shadow contributions). □

Remark 2.5 (Why the three invariants must be distinguished). A recurring error in expository passages is to conflate "shadow depth" (which means $r_{\max}$, the degree of the obstruction tower) with "collision depth" (which means $k_{\max}$, the pole order of the degree-2 collision residue) or with "maximal OPE pole order" (which means $p_{\max}$, the pole order in the generator self-OPE). These are three distinct invariants obeying the relation $k_{\max} = p_{\max} - 1$ but otherwise independent. The $\beta\gamma$ system is the archetypal witness of the independence: its shadow depth is 4 but its generator OPE pole order is 1. Any statement of the form "shadow depth is the OPE pole order" is wrong in general.

3 The seven faces

3.1 The master theorem

Theorem 3.1 (Seven-face identification)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra in the standard landscape, and let $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ be the collision residue of its universal Maurer–Cartan element. The following seven presentations all determine the same element of $\mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$:*

- **(F1)** Twisting morphism *(E_1 data): $r(z)$ is the genus-zero binary projection of the universal twisting morphism $\pi_{\mathcal{A}}\colon B(\mathcal{A}) \to \mathcal{A}$, where $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the ordered bar coalgebra with deconcatenation coproduct. The spectral parameter z survives because the ordered bar retains the collision coordinate; the Σ_n-coinvariant image loses it.*
- **(F2)** Line operator R-twist *(E_1 data): $r(z)$ is the R-matrix governing the meromorphic tensor product $V_{\ell_1} \otimes_{R(z_{12})} V_{\ell_2}$ of line operators in Dimofte–Niu–Py. The ordering of line operators on $\mathbb{R}$ is E_1-topological structure; the R-twist encodes the failure of commutativity.*
- **(F3)** Classical PVA r-matrix *(E_∞ shadow): $r(z)$ is the classical r-matrix extracted from the λ-bracket of the Poisson vertex algebra* $\mathrm{PVA}(\mathcal{A})$*. The λ-bracket is the semiclassical shadow of the chiral OPE; it retains the spectral parameter but not the ordering.*
- **(F4)** Gaiotto–Zeng commuting differentials *(E_1 data): the collision residue coefficients of $r(z)$ generate the commuting Hamiltonians $\{H_i\}$ on the sphere with n marked points. The Hamiltonians are ordered: they act on an n-fold ordered tensor product and their commutativity is the content of the Yang–Baxter equation at genus 0.*
- **(F5)** Yangian r-matrix *(E_1 data): $r(z)$ is the classical r-matrix of the Yangian $Y_\hbar(\mathcal{A}^!)$; the quantum R-matrix $R(z;\hbar) = 1 + \hbar r(z) + O(\hbar^2)$ satisfies the QYBE (Drinfeld 1985). The Yangian is the prototypical E_1-chiral algebra: its coproduct Δ_z depends on the spectral parameter.*
- **(F6)** Sklyanin bracket *(E_∞ shadow): $r(z)$ is the tensor generating the Sklyanin Poisson bracket on the space of $\mathcal{A}^!$-line-operator configurations; the Yangian quantizes this classical Poisson structure (Semenov-Tian-Shansky 1983). The Sklyanin bracket is the Poisson reduction of the R-matrix structure: it sees the spectral parameter but not the noncommutative coproduct.*
- **(F7)** Gaudin generator *(E_∞ shadow): $r(z)$ is the generating function of the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$ and their higher-collision generalizations (Feigin–Frenkel–Reshetikhin 1994). The Gaudin Hamiltonians commute as elements of $U(\mathfrak{g})^{\otimes n}$; the ordering is erased by the symmetry of the Casimir Ω.*

The identifications F1 $\Leftrightarrow$ F4, F1 $\Leftrightarrow$ F7, F1 $\Leftrightarrow$ F2, F1 $\Leftrightarrow$ F3 *are proved in this paper; the identification* F5 $\Leftrightarrow$ F6 *is Drinfeld's and Semenov-Tian-Shansky's classical result; the three-parameter normalization* $\hbar = 1/(k + h^\vee)$ *is proved here.*

3.2 Proof architecture

The seven-face identification is proved by a chain of six bilateral equivalences. Each equivalence is proved by an independent argument in a different categorical framework; the chain is closed by the identification of the universal MC element $\Theta_{\mathcal{A}}$ with the bar differential minus its linear part.

3.2.0.1 F1 $\Leftrightarrow$ F2. The twisting morphism $\pi_{\mathcal{A}}\colon B(\mathcal{A}) \to \mathcal{A}$ exists for every chirally Koszul chiral algebra by the bar-intrinsic construction $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. Its genus-zero binary projection is an element of $\mathcal{A} \otimes \mathcal{A}^![[z^{-1}]]$ (after the Koszul-dual coordinate change), which is the same as the R-matrix governing the meromorphic tensor product $V_{\ell_1} \otimes_{R(z_{12})} V_{\ell_2}$ of two line operators. The identification is proved in Volume II of the monograph under the label `thm:dnp-bar-cobar-identification`.

3.2.0.2 F1 $\Leftrightarrow$ F3. The Poisson vertex algebra $\mathrm{PVA}(\mathcal{A})$ is the semiclassical limit of $\mathcal{A}$, and the classical r-matrix extracted from the λ-bracket is the leading-order contribution to the quantum r-matrix. The identification with the collision residue of $\Theta_{\mathcal{A}}$ follows from the bar-intrinsic construction at genus 0:

the linear part d_0 encodes the PVA λ-bracket, and the collision residue extracts its z-dependence. The theorem label is `thm:kz-classical-quantum-bridge`.

3.2.0.3 F1 $\Leftrightarrow$ F4. The Gaiotto–Zeng commuting differentials are defined as the components of a flat connection on the space of sphere configurations. Flatness is equivalent to the commutativity $[H_i, H_j] = 0$, which follows from the Maurer–Cartan equation $d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ after projection to the collision-residue sector. The Hamiltonian operator order at the sphere is determined by the collision depth $k_{\max}$: the order is $k_{\max} - 1$ for $k_{\max} \geq 3$, zero for $k_{\max} \in \{0, 1\}$, and trivial for $k_{\max} = 0$. The theorem label is `thm:gz26-commuting-differentials`.

3.2.0.4 F4 $\Leftrightarrow$ F7. The Gaudin Hamiltonians are the $k = 1$ (simple-pole) truncation of the collision-residue expansion of $\Theta_{\mathcal{A}}$. For affine Kac–Moody at level k, the collision residue is $r(z) = \Omega/((k + h^\vee)z)$, where Ω is the Casimir and $(k + h^\vee)$ is the dual Coxeter shift. The Gaudin Hamiltonian is therefore $H_i^{\mathrm{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$, and the normalization relating it to the GZ26 Hamiltonian is $H_i^{\mathrm{GZ}} = (1/(k + h^\vee))H_i^{\mathrm{Gaudin}}$. The identification is proved under the label `thm:gaudin-yangian-identification`.

3.2.0.5 F5 $\Leftrightarrow$ F6. This is the classical theorem of Drinfeld (1985) and Semenov-Tian-Shansky (1983): the Yangian $Y_\hbar(\mathfrak{g})$ quantizes the classical r-matrix of the Sklyanin Poisson bracket. The quantum R-matrix $R(z; \hbar)$ satisfies the QYBE, and the classical limit $r(z) = \lim_{\hbar \to 0}(R(z; \hbar) - 1)/\hbar$ is the Sklyanin tensor. The new content of `thm:yangian-sklyanin-quantization` is the three-parameter normalization identifying $\hbar$ with the reciprocal of the level shift $(k + h^\vee)$.

3.2.0.6 Closing the chain. The chain F1–F2–F3–F4–F7–F5–F6 closes because F1 identifies $r(z)$ with the universal MC element $\Theta_{\mathcal{A}}$, and F6 identifies it with the Sklyanin tensor, and F7 identifies the Sklyanin tensor with the Gaudin generator (which is F4 via F7). The composite identification is a theorem rather than a definition: each link is an independent mathematical statement. □

3.3 The hinge: the bar-intrinsic construction

The primitive mathematical object underlying the seven-face programme is the E_1 Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ in the ordered convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$, where $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree tensor coalgebra with deconcatenation coproduct. The collision residue $r(z) = \mathrm{Res}_{0,2}^{\mathrm{coll}}(\Theta_{\mathcal{A}}^{E_1})$ retains the full spectral-parameter dependence; the modular MC element $\Theta_{\mathcal{A}} = \mathrm{av}(\Theta_{\mathcal{A}}^{E_1})$ and the modular characteristic are its Σ_n-coinvariant projections. Each of the seven faces is constructed from the ordered MC element; the modular shadow tower is the degree-by-degree av-image.

The entire programme rests on a single result: the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A}))$ exists and is determined by the bar differential of $\mathcal{A}$ via

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0,$$

where $D_{\mathcal{A}}$ is the full bar differential (incorporating all genera and all boundary strata) and d_0 is its linear part. The Maurer–Cartan equation follows from $D_{\mathcal{A}}^2 = 0$, which is a formal consequence of the boundary-of-boundary relation on $\overline{\mathcal{M}}_{g,n}$. This is Theorem 3.2 below; the proof is the content of the $\sim$ 2000-page Volume I and is not reproduced here.

Theorem 3.2 (The bar-intrinsic hinge). *For every chirally Koszul chiral algebra $\mathcal{A}$, the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is a Maurer–Cartan element of the cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$. Its genus-zero binary projection $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the collision residue, and the seven faces above are all proved from this single datum.*

3.4 The E_n circle and Arrow 2

The seven faces of $r(z)$ are Arrow 2 data in the E_n operadic circle

$$E_3(\text{bulk}) \xrightarrow{\text{codim-2 restriction}} E_2(\text{boundary chiral}) \xrightarrow{\text{ordered bar}} E_1(\text{bar/QG}) \xrightarrow{\text{Drinfeld center}} E_2 \xrightarrow{\text{derived center}} E_3.$$

The second arrow (ordered bar) extracts $r(z)$ from the E_2 boundary chiral algebra: the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A})$ is an E_1-coassociative coalgebra, and the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{E_1}_{\mathcal{A}})$ is the degree-2 datum of its Maurer–Cartan element. Faces F1, F2, F4, and F5 see the full E_1 structure (the spectral parameter, the noncommutative coproduct, the R-matrix); Faces F3, F6, and F7 see the E_∞ shadow (the Poisson bracket, the classical limit, the commuting Hamiltonians in the symmetric algebra). The averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ projects the spectral-parameter-dependent collision residue to the scalar shadow invariant κ: in abelian and scalar families this is $\mathrm{av}(r(z))$, while in non-abelian affine Kac–Moody it is $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$; the entire modular shadow tower is the Σ_n-coinvariant image of the ordered E_1 data.

The circle closes when the derived chiral center $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ (computed from the bar complex as a resolution) acquires E_3-topological structure via the topologization theorem (proved for affine Kac–Moody at non-critical level; conjectural in general). The seven faces live at the E_1 stage of this circle: they are the data extracted by the ordered bar before the E_2/E_3 reconstruction takes place.

4 The infinite face family: GRT-parametrized presentations

The master theorem of Section 3 identifies seven presentations. It is not the end of the story: each identification is implemented by a particular choice of associator relating two different coordinatizations of the same underlying E_1 datum. The Drinfeld Grothendieck–Teichmüller group $\mathrm{GRT}_1(\mathbb{Q})$ acts on the space of such coordinatizations by its tautological action on Drinfeld associators. The seven faces are seven orbit representatives; the action is free and the orbit is infinite.

4.1 The GRT-torsor of face presentations

Definition 4.1 (Face F_Φ). Let $\mathcal{A}$ be a chirally Koszul chiral algebra and let $\Phi \in \mathrm{GRT}_1(\mathbb{Q})$ be a Drinfeld associator valued in the prounipotent completion of the pure braid Lie algebra $\widehat{\mathfrak{t}}_n$. The Φ-*face* of $r(z)$ is

$$\mathrm{F}_\Phi(\mathcal{A}) \;:=\; \Phi_*\big(r_{\mathrm{bar}}(z)\big) \;\in\; \mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]],$$

where $r_{\mathrm{bar}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{E_1}_{\mathcal{A}})$ is the bar-intrinsic collision residue of Theorem 3.2 and Φ_* is the associator action on the $\widehat{\mathfrak{t}}_2$-valued genus-zero binary datum.

Theorem 4.2 (GRT-torsor of face presentations). *For every chirally Koszul chiral algebra $\mathcal{A}$ in the standard landscape:*

(i) *The assignment* $\Phi \mapsto F_\Phi(\mathcal{A})$ *is a free transitive action of* $\mathrm{GRT}_1(\mathbb{Q})$ *on the set of coordinatized presentations of the universal collision residue.*

(ii) *The bar-intrinsic face* F_1 *(Definition* 4.1 *with* $\Phi = 1$ *the trivial associator) is the origin of the torsor.*

(iii) *The seven listed faces* $\{F_i\}_{i=1}^7$ *of Theorem* 3.1 *are values of* F_{Φ_i} *at seven explicit associators* $\Phi_1, \ldots, \Phi_7 \in \mathrm{GRT}_1(\mathbb{Q})$ *listed in Table* 4.7 *below.*

(iv) *Any two faces* $F_\Phi, F_{\Phi'}$ *differ by the* GRT*-element* $\Phi'\Phi^{-1}$ *acting on the shared underlying* $\widehat{\mathfrak{t}}_2$*-datum. Equality of presentations (as elements of* $\mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$*, not as syntactic formulas) is the statement* $F_\Phi = F_{\Phi'}$ *at the level of the torsor quotient.*

(v) *The* GRT*-action factors through the finite-dimensional abelianization* $\mathrm{GRT}^{\mathrm{fin}} = \mathrm{GRT}_1/[\mathrm{GRT}_1, \mathrm{GRT}_1]$ *on the genus-zero binary datum. On affine Kac–Moody, the image is the one-parameter Casimir rescaling subgroup* $r(z) \mapsto \lambda \cdot r(z)$ *realising the conversion* $k\,\Omega/z \rightleftharpoons \Omega/((k+h^\vee)z)$ *as the rescaling by* $\lambda = 1/(k+h^\vee)$ *(a tensor identity, not a chained equality; see Section 4.4).*

Sketch. (i) The space of coordinatized presentations is by construction a pseudo-torsor under GRT_1, because two presentations differ by a gauge of the braid Lie datum and the gauge group is exactly GRT_1 (Drinfeld, "On quasitriangular quasi-Hopf algebras and a group closely connected with $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$"). Transitivity is the Drinfeld–Kohno theorem relating two associators by an inner automorphism. Freeness follows from the fact that the bar-intrinsic presentation $r_{\mathrm{bar}}(z)$ has only the identity stabilizer in the associator gauge group.

(ii) is the bar-intrinsic hinge: $F_1(\mathcal{A}) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is produced by the genus-zero binary projection of the universal MC element with no further coordinate choice.

(iii) is verified face by face in Table 4.7.

(iv) is the cocycle relation $\Phi'\Phi^{-1} \cdot F_\Phi = F_{\Phi'}$ for any torsor action.

(v) is the Furusho degeneration: on the genus-zero binary datum, the higher associator terms in $\widehat{\mathfrak{t}}_n$ for $n \geq 3$ project to zero, and the residual action is generated by the linear rescaling of the pure braid generator t_{12}. For affine Kac–Moody, $t_{12} \mapsto \Omega$ is the Casimir and the rescaling parameter λ is the inverse level shift. □ □

Remark 4.3 (Why this is an upgrade, not a relabeling). Theorem 4.2 says the seven faces are not a finite list: they are seven explicit choices of associator gauge, and any other associator Φ produces a face F_Φ which is equally valid. Brown's motivic associator Φ^{mot} and Willwacher's graph-complex associator Φ^{Will} give two further natural faces F_8, F_9 (Sections 4.2 and 4.3). The framework is open-ended: every new associator construction produces a new face, and coherence with the existing seven is automatic (it is the torsor axiom), not a separate theorem to be proved.

4.2 The motivic face F_8

Theorem 4.4 (Motivic face). *Let* $\Phi^{\mathrm{mot}} \in \mathrm{GRT}_1(\mathbb{Q})$ *be the canonical motivic associator of Brown (*Mixed Tate motives over $\mathbb{Z}$*, Annals of Math. 2012). Applied to the bar-intrinsic collision residue* $r_{\mathrm{bar}}(z)$*, it produces the* motivic face

$$F_8(\mathcal{A})(z) \;=\; \sum_{n\geq 1} \frac{r_n(\mathcal{A})}{z^n} \;+\; \sum_{w\geq 3} \sum_{n_1,\ldots,n_r} \zeta(n_1,\ldots,n_r)\, \tau_{n_1,\ldots,n_r}(\mathcal{A}) \cdot z^{-w},$$

where $\zeta(n_1, \ldots, n_r)$ is a multiple zeta value of weight $w = n_1 + \cdots + n_r$ and $\tau_{n_1,\ldots,n_r}(\mathcal{A})$ is an explicit rational combination of iterated OPE brackets on $\mathcal{A}$. The MZV coefficients appear at weight ≥ 3; at weights 1 and 2 the motivic face agrees with the bar-intrinsic face.

Sketch. The motivic associator Φ^{mot} is defined by the Drinfeld–Kontsevich integral on $\mathcal{M}_{0,4} \cong \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and its coefficients are the canonical period integrals of the motive $\mathbb{H}^1(\mathbb{P}^1 \setminus \{0, 1, \infty\})$, which are exactly the motivic multiple zeta values (Brown 2012). Applying Φ^{mot}_* to the $\widehat{\mathfrak{t}}_2$-presentation of $r_{\mathrm{bar}}(z)$ produces an expansion in t_{12}^n with MZV coefficients at weight ≥ 3. The coefficient $\tau_{n_1,\ldots,n_r}$ is extracted by composing the iterated OPE bracket of depth r with the motivic comodule action. The $\mathrm{GRT}^{\mathrm{fin}}$-projection kills the MZV terms (they lie in the commutator subalgebra), so for algebras where the face identity holds through the abelianization (affine Kac–Moody, Heisenberg) the motivic face agrees with the bar face; for class M algebras (Virasoro, $\mathcal{W}_N$) where the higher commutator terms are detected, F_8 carries genuinely new MZV data. □

4.3 The operadic face F_9

Theorem 4.5 (Operadic face)**.** *Let GC_2 be the Kontsevich graph complex of stable 2-dimensional graphs, and let $\mathrm{GRT}_1(\mathbb{Q}) \cong H^0(\mathrm{GC}_2)$ be Willwacher's isomorphism (*Grothendieck–Teichmüller group and Kontsevich graph complex*, Invent. Math. 2015). For every graph cocycle $\gamma \in \mathrm{GC}_2$, the corresponding* GRT*-element Φ^γ acts on the bar complex $B(\mathcal{A})$ by a graph-summed derivation. The* operadic face *is*

$$\mathrm{F}_9(\mathcal{A})(z) \;=\; \sum_{\Gamma} \frac{1}{|\mathrm{Aut}\,\Gamma|}\, c_\Gamma(z) \cdot W_\Gamma(\mathcal{A}),$$

summed over stable graphs Γ with two external legs, where $c_\Gamma(z)$ is the graph weight (a rational function of z computed from Kontsevich configuration integrals) and $W_\Gamma(\mathcal{A})$ is the corresponding multi-OPE bracket on $\mathcal{A}$. The face F_9 is obtained from F_1 by the GRT*-element dual to the graph cocycle.*

Remark 4.6 (Operadic interpretation)*.* Face F_9 realizes $r(z)$ as a graph-sum over Feynman diagrams in the Kontsevich formality quasi-isomorphism for the E_2-operad. It is the face that sees the operadic (topological configuration-space) origin of the collision residue most directly: each stable graph contributes a universal rational function $c_\Gamma(z)$, and the sum over graphs is controlled by the graph-complex cohomology. In contrast to F_1 (bar-intrinsic: coordinate-dependent but canonical) and F_8 (motivic: period-valued), F_9 is manifestly configuration-space-geometric.

4.4 The landscape census reconciliation

The Vol I landscape census lists the affine Kac–Moody collision residue in two forms that appear contradictory:

$$r(z) \;=\; \frac{k\,\Omega}{z} \qquad \text{and} \qquad r(z) \;=\; \frac{\Omega}{(k + h^\vee)\,z}.$$

These are *not* a chained equality (as though $k = 1/(k + h^\vee)$, which is false). They are *two faces* in the $\mathrm{GRT}^{\mathrm{fin}}$-torsor of Theorem 4.2(v), related by the Casimir rescaling $\lambda = 1/(k(k + h^\vee))$:

$$\Phi^{\mathrm{KM}}_{k \to 1/(k+h^\vee)} \cdot \big(k\,\Omega\big) \;=\; \frac{\Omega}{k + h^\vee}.$$

The left-hand side is the bar-intrinsic (unshifted) face produced by the ordered bar propagator with propagator normalization $1/z$; the right-hand side is the Sugawara-normalized face produced by absorbing the level shift into the Casimir tensor. Both present the same underlying element; the equality is a *tensor identity in the* $\mathrm{GRT}^{\mathrm{fin}}$*-quotient*, not a numerical identity of coefficients.

Remark 4.7 (How this resolves). The failure mode recorded the attempted chain $k\,\Omega/z = \Omega/((k+h^\vee)z)$ as a numerical identity and found it false (at $k=1$, $h^\vee=2$ for $\mathfrak{sl}_2$: $1\cdot\Omega = \Omega/3$ is false as scalars). The reconstitution is: these are two orbit representatives, and the GRT-torsor is exactly the group that intertwines them. (level-stripped r-matrix) is compatible with this reconstitution: both faces satisfy $k=0 \Rightarrow r=0$ (the left face vanishes because its numerator is $k\,\Omega$; the right face vanishes because the Sugawara normalization is undefined when $k+h^\vee=\infty$ at $k=0$ via the reciprocal).

4.5 The F_7 disambiguation and the super-variant

Two distinct structures have been called "face F_7" in the literature and in earlier drafts:

- $\mathrm{F}_7^{\mathrm{Gaudin}}$: the simple-pole projection r_1 (the generating function of the Gaudin Hamiltonians $H_i = \sum_{j\neq i}\Omega_{ij}/(z_i - z_j)$). This lives on the E_∞ shadow and is defined for all four shadow classes.
- $\mathrm{F}_7^{M,\mathrm{top}}$: the top-degree A_∞ operation m_3 detecting non-formality (the class-M-specific face). This lives on the E_1 datum itself and is nonzero only for class M (Virasoro, $\mathcal{W}_N$).

Proposition 4.8 (F_7 disambiguation). *The Gaudin generator* $\mathrm{F}_7^{\mathrm{Gaudin}}$ *is the simple-pole coefficient* r_1 *of the collision expansion* $r(z) = \sum_{k\geq 0} r_k\, z^{-k-1}$*; it is* $\mathrm{GRT}^{\mathrm{fin}}$*-invariant (a scalar per Casimir rescaling) and is common to all four classes* G, L, C, M*. The top-*A_∞ *face* $\mathrm{F}_7^{M,\mathrm{top}}$ *is the class-*M *extension: the operation* $m_3\colon \mathcal{A}^{\otimes 3} \to \mathcal{A}$ *that encodes the non-formality of the bar complex at the highest collision depth.* $\mathrm{F}_7^{M,\mathrm{top}}$ *is vacuous for classes* G, L, C *(where* $m_3 = 0$ *by formality) and carries genuine new data for class* M*.*

The naming is: F_7 without superscript means $\mathrm{F}_7^{\mathrm{Gaudin}}$ (the default); the top-A_∞ face is always written $\mathrm{F}_7^{M,\mathrm{top}}$ or equivalently F_7'.

4.6 The super-variant

For chiral algebras with *odd* generating fields (symplectic fermions, bc ghosts, super-affine Kac–Moody), the collision residue from simple poles is distinct from the Laplace-transform r-matrix. The bar-intrinsic construction is unchanged, but the $\mathrm{F}_1 \Leftrightarrow \mathrm{F}_2$ identification requires a sign twist.

Theorem 4.9 (Super-variant of Theorem 3.1). *Let* $\mathcal{A}$ *be a super-chirally-Koszul chiral superalgebra (generating fields graded by* $\mathbb{Z}/2$*). All statements of Theorems* 3.1 *and* 4.2 *hold verbatim after the following substitutions:*

(a) *The tensor product* $\otimes$ *is the* $\mathbb{Z}/2$*-graded (Koszul-sign) tensor product.*

(b) *The Casimir* Ω *is the super-Casimir* $\Omega^{\mathrm{super}} = \sum_a (-1)^{|a|} t_a \otimes t^a$*, computed using the supertrace-invariant bilinear form.*

(c) *The associator gauge group is the super-*GRT_1 *acting on the parity-graded pure braid Lie superalgebra* $\hat{\mathfrak{t}}_n^{\mathrm{super}}$.

(d) *The* $\mathrm{F}_1 \Leftrightarrow \mathrm{F}_2$ *identification is mediated by the super-twist* $\sigma_{\mathrm{super}}(a\otimes b) = (-1)^{|a||b|}\, b\otimes a$ *rather than the bosonic flip.*

The $k_{\max} = p_{\max} - 1$ relation, the operator-order trichotomy, and the four-class classification are unchanged. The $\beta\gamma$ system and the symplectic fermion example are the two archetypal class-C super-landscape witnesses.

4.7 Table of representatives

Face	Associator Φ	Gauge	Where this face is native
F_1	1 (trivial)	bar-intrinsic	universal (Vol I Thm A)
F_2	Φ^{DNP}	line-operator R-twist	3d HT (DNP 2025)
F_3	$\Phi^{\mathrm{KZ-PVA}}$	λ-bracket	PVA (Khan–Zeng 2025)
F_4	Φ^{GZ26}	sphere Hamiltonian	holographic (GZ 2026)
F_5	Φ^{KZ} (Knizhnik–Zamolodchikov)	Yangian $\hbar$	Yangian (Drinfeld 1985)
F_6	Φ^{STS}	Sklyanin flip	Poisson (STS 1983)
F_7	Φ^{FFR}	Casimir simple pole	Gaudin (FFR 1994)
$\mathrm{F}_7^{M,\mathrm{top}}$	–- (not GRT-conjugate)	class-M A_∞ top	Virasoro, $\mathcal{W}_N$ (this work)
F_8	Φ^{mot} (Brown motivic)	MZV-valued	mixed Tate motives (Brown 2012)
F_9	Φ^{Will} (Willwacher graph)	Kontsevich graph complex	E_2-formality (Willwacher 2015)
F_Φ	any $\Phi \in \mathrm{GRT}_1(\mathbb{Q})$	torsor orbit	general associator gauge

The first seven rows are the historical faces of Theorem 3.1. Rows 8–9 are the GRT-reconstitution faces of Theorems 4.4 and 4.5. The bottom row is the general member of the GRT-orbit; the face family is infinite and parametrized by $\mathrm{GRT}_1(\mathbb{Q})$. The row $\mathrm{F}_7^{M,\mathrm{top}}$ is *not* in the GRT-orbit of F_1: it is an independent face detecting non-formality in class M only, and the torsor identification does not apply to it.

5 Chiral quantum group equivalence on the ordered Koszul locus

The seven-face hub routes through a pentagon of structure-equivalences (Section 4 is the star shape; this section records the load-bearing diagonal). The single hinge is the chiral quantum group equivalence on the ordered Koszul locus: three structures (classical r-matrix, A_∞-chiral coproduct, ordered bar coproduct with spectral parameter) determine each other there.

Definition 5.1 (Ordered Koszul chiral algebra)**.** A chiral algebra $\mathcal{A}$ on X is *ordered-Koszul* if

(i) the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is concentrated in weight 1 on $H^\bullet$ (PBW collapse); and

(ii) the pure-braid Orlik–Solomon form factor identifies $H^\bullet(B^{\mathrm{ord}}(\mathcal{A}))$ with $\mathrm{OS}(\mathfrak{t}_n) \otimes \mathcal{A}^{\otimes n}$ up to Σ_n-twist, lifting the twisted-tensor witness of spoke 4 of the six-fold Koszul-locus TFAE to the ordered setting.

The affine Yangian $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ of Remark 5.2 is ordered-Koszul via Drinfeld's second presentation combined with Etingof–Kazhdan flatness.

Remark 5.2 (Four Yangian types must be kept distinct). We distinguish four Yangian notions and mark each with a superscript to avoid type errors across the programme:

(1) $Y_\hbar(\mathfrak{g})$ *classical*: the E_1-topological Yangian on $\mathbb{R}$ in Drinfeld's first presentation.

(2) $Y_\hbar^{\mathrm{dg}}(\mathfrak{g})$ *dg-shifted*: the derived Yangian at a point or on the formal disk; see Section **??** of the companion.

(3) $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ *chiral*: E_1-chiral on the curve X, presented as a D-module with spectral parameter living in the ordered configuration coordinate.

(4) $Y_\hbar(\mathfrak{g})^{\mathrm{spec}}$ *spectral*: the factorization algebra on $\mathbb{A}^1_u$ obtained by transposing the spectral coordinate.

Conflating any two of these four incarnations (classical on $\mathbb{R}$; dg-shifted on the formal disk; chiral on the curve X; spectral on $\mathbb{A}^1_u$) is a type error: each carries its own derived centre and its own notion of R-matrix. The chiral QG equivalence below concerns type (3).

Theorem 5.3 (Chiral QG equivalence on the ordered Koszul locus). *Let $\mathcal{A}$ be ordered-Koszul in the sense of Definition 5.1. The following three structures on $\mathcal{A}$ are equivalent and mutually determining:*

(A) *a classical r-matrix $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$ satisfying CYBE;*

(B) *an A_∞-chiral coproduct $\Delta^{\mathrm{ch}} = \{\Delta_k\}_{k\geq 2}$ on $\mathcal{A}$ with modular MC tower;*

(C) *an ordered bar coproduct on $B^{\mathrm{ord}}(\mathcal{A})$ carrying deconcatenation and a spectral parameter compatible with the Kohno connection.*

The equivalences $(A) \Leftrightarrow (B) \Leftrightarrow (C)$ hold up to an associator gauge $\Phi \in \mathrm{GRT}_1(\mathbb{Q})$: the cochain-level identification is associator-dependent, whereas the bar side is associator-free on H^0 by Theorem 5.5.

Remark 5.4 (Scope of the chiral QG equivalence). Theorem 5.3 is abstract on the ordered Koszul locus; concrete verifications are currently fully inscribed for: $Y_\hbar(\mathfrak{sl}_2)^{\mathrm{ch}}$ (the Drinfeld second-presentation Yangian), affine $\widehat{\mathfrak{g}}_k$ at non-critical k, and the elliptic and toroidal formal-disk instances of Sections **??** and **??**. General-type $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ falls under the abstract Koszul-locus umbrella but awaits a concrete RTT–to–Drinfeld bridge outside type A_1.

Theorem 5.5 (GRT_1 rigidity on the bar side). *The bar-side invariants of $\mathcal{A}$ –- κ, the shadow tower $\{S_r\}$, and the cohomology-level Maurer–Cartan datum $[\Theta_{\mathcal{A}}]$ –- are $\mathrm{GRT}_1(\mathbb{Q})$-trivial on H^0: any two associators $\Phi, \Phi' \in \mathrm{GRT}_1(\mathbb{Q})$ produce equal bar-cohomology-level data. At the cochain level the action is homotopic but not trivial: associator-dependence concentrates in the coproduct/cobar side, not in the bar shadow.*

Conjecture 5.6 (Elliptic chiral QG equivalence leading order $\hbar$). *For $\mathfrak{sl}_2$ on a smooth elliptic curve E_τ, the Belavin non-dynamical elliptic r-matrix $r^{\mathrm{ell}}_{\mathfrak{sl}_2}(z,\tau)$, constructed from the full Weierstrass zeta $\zeta_\tau(z) = \theta_1'(z|\tau)/\theta_1(z|\tau) + 2\eta_1 z$ in the Cartan channel and from the theta-function ratios $\phi_\pm(z,\tau)$ in the root channels, is the classical shadow of a chiral quantum group on E_τ. The quasi-period correction $2\eta_1 z$ is not optional: dropping it and writing only the theta derivative θ_1'/θ_1 breaks the classical Yang–Baxter equation off the diagonal (this is the bug warned against in the genus-1 convention note; the correct Belavin form carries the full ζ_τ). At this level the Belavin r-matrix and the Knizhnik–Zamolodchikov–Bernard connection determine each other via elliptic regularization of the genus-*0 *collision expansion (chapter-level: Vol I* prop:elliptic-rmatrix-shadow *and Vol I* thm:g1sf-elliptic-rmatrix*, both for the Belavin side).*

The Felder dynamical *R-matrix $R(z,\lambda,\tau)$, which satisfies the dynamical YBE rather than the ordinary CYBE, is a distinct IRF/face-model cousin. At $\mathfrak{sl}_2$ the vertex–IRF correspondence of Felder–Pasquier relates*

Belavin non-dynamical to Felder dynamical via a gauge transform; the equivalence to KZB at leading order in $\hbar$ and the role of the Fay trisecant identity (equivalently the Szegő three-term identity) as the modular substitute for the Drinfeld pentagon are inscribed only in the present standalone, not yet in a chapter, and therefore carry status pending chapter-level inscription. The heat-equation prefactor $1/(4\pi i)$ on the diagonal and $1/(2\pi i)$ off-diagonal (cf. the genus-1 convention note on the KZB heat prefactor) is part of the conjectural Felder–KZB data.

Theorem 5.7 (Toroidal chiral QG equivalence on the formal disk)**.** *The Ding–Iohara–Miki toroidal algebra $U_{q,t}(\ddot{\mathfrak{sl}}_N)$ with its two spectral parameters (u, v) is the chiral quantum group of an ordered-Koszul factorization algebra on the formal disk* $(\operatorname{Spec}\mathbb{C}[[z]])^{\times 2}$*; the equivalence is witnessed concretely via the Schiffmann–Vasserot cohomological Hall algebra on the instanton moduli space, with the Miki S_3 automorphism of the three Ω-background parameters (q_1, q_2, q_3) under $q_1q_2q_3 = 1$ acting as the triality of the toroidal presentation. Global extension to $\mathbb{P}^1 \times \mathbb{P}^1$ is conditional on the class-M chain-level topologization frontier on the* original *bar complex (direction (I) of the topologization-tower open inventory: the weight-completed, pro-object, and J-adic ambients already yield chain-level class-M topologization; the frontier is the* $\mathrm{Ch}(\mathrm{Vect})$ *direct-sum ambient on the raw complex).*

Remark 5.8 (Scope and attribution). The formal-disk inscription lives in Vol III `chapters/theory/quantum_groups_foundatio` (proposition "Algebra side of toroidal Kazhdan–Lusztig, $\mathfrak{sl}_2$, formal disk"), extended to $\mathfrak{sl}_N$ via the Feigin–Hashizume–Hoshino–Shiraishi–Yanagida shuffle presentation. The Drinfeld–Miki name is standard usage in the quantum-toroidal literature; the Ginzburg connection appears via the Nakajima quiver-variety construction of the Schiffmann–Vasserot CoHA. On the formal disk the equivalence is bialgebra-level (AP263: the vertex-algebraic antipode lift is a separately proved negative for toroidal just as for Yangian), and the "quantum group" label here is shorthand for "chiral bialgebra in the sense of **??**". The global $\mathbb{P}^1 \times \mathbb{P}^1$ obstruction is the identical class-M original-complex direction as the Virasoro/$\mathcal{W}_N$ topologization frontier; closing that frontier closes the toroidal global case as a downstream corollary.

Remark 5.9 (Pentagon of equivalences routed through the operadic hub). The five presentations of $\mathrm{SC}^{\mathrm{ch,top}}$ (operadic, Koszul, factorization, BV–BRST, convolution) form a *star*, not a pentagon: direct pairwise bridges hold only for Koszul–BV, BV–Convolution, and Factorization–Convolution; the remaining two arrows factor through the operadic hub. This star topology is the correct combinatorial picture behind the seven-face diagram as well: the bar-intrinsic face F_1 is the hub, and the six historical faces F_2–F_7 are spokes routed through it.

5.1 Miura cross-universality at all spins

The coefficient appearing in chiral coproducts on the cross-terms $J \otimes W_{s-1} + W_{s-1} \otimes J$ is universal in the spin s.

Theorem 5.10 (Miura cross-universality at all spins)**.** *Let $\mathcal{W}_N$ be the principal W-algebra at level parameter $\Psi = (k+N)/(k+N-1)$ (with $\Psi \neq 0, 1$ the generic domain). For every spin $s \geq 2$, the Drinfeld coproduct on the primary generator W_s of $\mathcal{W}_N$ satisfies*

$$\Delta_z(W_s) \;=\; W_s \otimes 1 + 1 \otimes W_s \;+\; \frac{\Psi - 1}{\Psi}\left(J \otimes W_{s-1} + W_{s-1} \otimes J\right) \;+\; \textit{composite corrections},$$

where the composite corrections are normally-ordered products of lower-spin generators vanishing at $\Psi = 1$. The cross-term coefficient $(\Psi-1)/\Psi$ is universal*: it is independent of N and of s for all $s \geq 2$.*

Remark 5.11 (What the theorem retracts). Earlier drafts recorded the cross-term coefficient as $1/\Upsilon$. At $\Upsilon = 2$ the two coefficients agree numerically ($(\Upsilon - 1)/\Upsilon = 1/\Upsilon = 1/2$): this coincidental agreement masked the error. Substituting $\Upsilon = 3$ separates the candidates, since $(\Upsilon - 1)/\Upsilon = 2/3$ but $1/\Upsilon = 1/3$; the universal coefficient is $(\Upsilon - 1)/\Upsilon$.

Remark 5.12 (Degenerate loci $\Upsilon = 0, 1$). At $\Upsilon = 1$ (the free-boson point), the cross-term vanishes and Δ_z collapses to the plain coproduct. At $\Upsilon = 0$ (the free-abelian degeneration, $k = -N$), the cross-term coefficient diverges as a rational function; the correct statement is that the chiral algebra degenerates to an abelian factor with identity quantum group structure, and the Miura coefficient is not defined. These two degenerations are the endpoints of the Yangian quantization line.

5.2 $\mathfrak{gl}_N$ chiral QG via Feigin–Frenkel screening

The JKL26 dependence in earlier drafts of the $\mathfrak{gl}_N$ chiral quantum group has been eliminated: the programme is unconditional for all $N \geq 2$ via Feigin–Frenkel screening.

Theorem 5.13 ($\mathfrak{gl}_N$ chiral QG, unconditional all $N \geq 2$)**.** *Let $\mathcal{W}_N$ be the principal W-algebra of $\mathfrak{gl}_N$ at $\Upsilon = (k + N)/(k + N - 1)$, realised as the intersection of screening operator kernels*

$$\mathcal{W}_N = \bigcap_{i=1}^{N-1} \ker(Q_{\alpha_i}) \quad \subset \quad \mathcal{H}_{N-1},$$

inside the rank-$(N-1)$ Heisenberg vertex algebra $\mathcal{H}_{N-1}$ (class G). The Drinfeld coproduct on $\mathcal{H}_{N-1}$ descends to the intersection, and the RTT relations on the resulting chiral quantum group are inherited from the abelian parent. The two-parameter $\mathfrak{gl}_N$ presentation matches the $\mathrm{Sym}^2(\mathfrak{g}^*)^{\mathfrak{g}} = 2$-dimensional space of invariant bilinear forms on $\mathfrak{gl}_N$ (trace B_{tr} and abelian B_{ab}), giving two independent quantum deformation parameters.

Remark 5.14 (Hook-type non-principal W extension). For non-principal nilpotents of hook type $(r, 1^{N-r})$ with $r \leq N - 3$, the same screening presentation goes through via the parabolic screenings of Kac–Roan–Wakimoto and the affine extension of Arakawa. Subregular and minimal W-algebras reduce to $\mathcal{W}_{N-2} \times \widehat{\mathfrak{sl}}_2 \times \beta\gamma$ by the class-C SDR; this is genuine residual mathematics rather than a plumbing obstruction.

6 The classification trichotomy

6.1 The four shadow classes

The single-line dichotomy partitions the standard landscape into four classes, classified by the shadow depth $r_{\max}$.

Class	$r_{\max}$	β_{level}	**Examples**
G (Gaussian)	2	quadratic	Heisenberg, lattice VOAs, free fermion, bc ghosts
L (Lie/tree)	3	quadratic	Affine $\widehat{\mathfrak{g}}_k$ all simple types, non-critical
C (Contact)	4	simple-pole	$\beta\gamma$, symplectic fermions, matrix factorizations
M (Mixed)	∞	higher-pole	Virasoro, $\mathcal{W}_N$, principal DS reductions

The classes are mutually exclusive and exhaustive. Every chirally Koszul algebra in the standard landscape falls into exactly one class, and every class is realized by at least one family.

6.2 The three invariants table

For each family in the standard landscape, the triple $(p_{\max}, k_{\max}, r_{\max})$ takes an explicit value that determines the class and the operator order of the GZ26 Hamiltonians.

Algebra	$p_{\max}$	$k_{\max}$	$r_{\max}$	**Op. order**	**Class**
Heisenberg $\mathcal{H}_k$	2	1	2	0	G
Lattice V_Λ	2	1	2	0	G
Affine $\widehat{\mathfrak{sl}}_{N\,k}$	2	1	3	0	L
$\beta\gamma$	1	0	4	trivial	C
Virasoro Vir_c	4	3	∞	2	M
$\mathcal{W}_3$	6	5	∞	4	M
$\mathcal{W}_N$	$2N$	$2N-1$	∞	$2N-2$	M
Bershadsky–Polyakov $\mathcal{W}_3^{(2)}$	4	3	∞	2	$M\,(T)\,/\,G\,(J)$

6.3 The operator-order trichotomy

Theorem 6.1 (Operator-order trichotomy)**.** *For every $\mathcal{A}$ in the standard landscape, the collision depth $k_{\max}(\mathcal{A}) \in \{0, 1\} \cup \{3, 4, 5, \ldots\}$. The classification into three operator regimes is:*

(i) Trivial*: $k_{\max} = 0$. The GZ26 Hamiltonians are identically zero. Examples: $\beta\gamma$, symplectic fermions (class C).*

(ii) Multiplication*: $k_{\max} = 1$. The GZ26 Hamiltonians are multiplication operators of order* 0*. Examples: Heisenberg (class G), affine Kac–Moody (class L).*

(iii) Differential*: $k_{\max} \geq 3$. The GZ26 Hamiltonians are genuine differential operators of order $k_{\max} - 1$. Examples: Virasoro (order 2), $\mathcal{W}_N$ (order $2N-2$); both are class M.*

The value $k_{\max} = 2$ does not occur in the standard landscape, because it would require $p_{\max} = 3$, which is forbidden by bosonic locality and half-integer weight constraints.

Proof. The exclusion of $k_{\max} = 2$ is Proposition 6.2 below. The three cases are:

- $k_{\max} = 0$: the collision residue has only the constant (z^0) term, which by degree counting cannot produce a nontrivial differential operator at any order; therefore the Hamiltonians vanish.
- $k_{\max} = 1$: the collision residue has a simple pole at $z = 0$, which by residue extraction gives a multiplication operator. For affine Kac–Moody, this is the Casimir.
- $k_{\max} \geq 3$: the collision residue has poles of order ≥ 3, whose residue extraction gives differential operators of positive order. For Virasoro at $k_{\max} = 3$, the order is 2; for $\mathcal{W}_N$ at $k_{\max} = 2N - 1$, the order is $2N - 2$.

The match with the four-class shadow classification is direct: classes G and L both have $k_{\max} = 1$; class C has $k_{\max} = 0$; class M has $k_{\max} \geq 3$. □

Proposition 6.2 (Absence of $k_{\max} = 2$). *The value $k_{\max} = 2$ does not occur in the standard landscape.*

Proof. The generator OPE pole order $p_{\max}$ takes values in $\{1, 2\} \cup \{4, 5, \ldots\}$ for bosonic chiral algebras in the standard landscape. The OPE of two weight-h currents has poles at orders $0, 1, \ldots, 2h$, and the maximal nonzero pole is $2h$ only for bosonic self-OPEs. For $h = 1$ (currents), $p_{\max} = 2$; for $h = 2$ (stress tensors), $p_{\max} = 4$; intermediate values of $p_{\max} = 3$ would require half-integer weight, which breaks bosonic locality. Since $k_{\max} = p_{\max} - 1$, the corresponding excluded value is $k_{\max} = 2$. □

6.4 Why $\beta\gamma$ is exceptional

The $\beta\gamma$ system is the critical witness for the independence of $r_{\max}$ from $p_{\max}$. Its shadow depth is $r_{\max} = 4$ (class C), but its generator OPE pole order is $p_{\max} = 1$ (not 2, not 4). The reason: the generators β and γ have the simple-pole OPE $\beta(z)\gamma(w) \sim (z - w)^{-1}$, with $\beta\beta$ and $\gamma\gamma$ both regular. The Sugawara energy-momentum tensor $T =: (1 - \lambda)\beta\partial\gamma - \lambda\partial\beta\gamma :$ is a composite field of weight 2 whose self-OPE has a quartic pole $(z - w)^{-4}$, but T is *not* a generator: it is determined by β and γ and does not appear in the generating set.

The consequence for the seven-face programme is that the GZ26 commuting Hamiltonians of the $\beta\gamma$ system vanish identically: there is nothing to quantize at the sphere level, because the collision residue $r(z)$ of $\Theta_{\beta\gamma}$ is zero. The shadow depth $r_{\max} = 4$ of $\beta\gamma$ comes not from higher OPE poles but from the quartic contact invariant, which lives on a charged stratum of the cyclic deformation complex and exits the complex by rank-one rigidity (stratum separation).

Remark 6.3 (Stratum separation). Class C is characterized by a mechanism distinct from classes G, L, and M: the termination of the shadow obstruction tower at degree 4 occurs not because the shadow metric Q_L becomes a perfect square (as in G and L) nor because it remains infinite (as in M), but because the quartic contact invariant lives on a stratum whose self-bracket exits the complex. On the primary line, Q_L is the constant $(2\kappa)^2$, but the full cyclic deformation complex has additional charged strata that carry the contact invariant. This is why class C appears in the classification with the annotation $\Delta = 0^*$: the discriminant vanishes on the primary line but the contact obstruction is nonzero in the full complex.

7 Compute engines

The seven-face programme is backed by fourteen compute engines, each independently verifying one or more face identifications. Every engine has a test file with multiple independent verification paths per tested formula.

Engine	Tests	Faces
theorem_gz26_commuting_hamiltonians_engine	32	F4, F7
theorem_dnp_meromorphic_tensor_engine	23	F1, F2
theorem_bv_brst_genus1_constraints_engine	59	F3 (genus 1)
theorem_w3_holographic_datum_engine	83	F4, F7 ($\mathcal{W}_3$)
theorem_pva_classical_r_matrix_engine	61	F3, F5
theorem_virasoro_spectral_r_matrix_engine	91	F5 (Vir)
theorem_nonprincipal_line_operators_engine	63	F2 (non-principal DS)
theorem_genus2_planted_forest_gz26_engine	73	F4 (genus 2)
theorem_w3_commuting_hamiltonians_engine	105	F4, F7 ($\mathcal{W}_3$)
theorem_three_way_r_matrix_engine	72	F5, F6, F7
theorem_ainfty_nonformality_class_m_engine	77	trichotomy (M)
theorem_dk0_evaluation_bridge_engine	63	F2, F4
theorem_sl3_yangian_r_matrix_engine	88	F5 ($\mathfrak{sl}_3$)
theorem_three_paper_intersection_engine	39	invariants, trichotomy
Total	**929**	

The critical engine is `theorem_three_paper_intersection_engine`: its 39 tests check the $(p_{\max}, k_{\max}, r_{\max})$ triple for every row of the three-invariants table and confirm the operator-order assignment. The $\beta\gamma$ row is verified by the dedicated test `test_betagamma_class_C`, which is the sharpest cross-check for the three-invariants distinction: it confirms $p_{\max}(\beta\gamma) = 1$, $k_{\max}(\beta\gamma) = 0$, $r_{\max}(\beta\gamma) = 4$, operator order trivial, class C.

8 Status and loose ends

8.1 Volume status

The seven-face programme is proved for the standard landscape in the three-volume monograph. The per-face per-volume status is:

Face	Vol I	Vol II	Vol III	Source
F1 (twisting morphism)	proved	specialized	specialized	Vol I Thm A
F2 (line operator R-twist)	cited	proved	specialized	DNP 2025
F3 (PVA r-matrix)	proved	used	specialized	KZ 2025
F4 (GZ26 commuting)	proved	used	specialized	GZ 2026
F5 (Yangian r-matrix)	proved	cited	specialized	Drinfeld 1985
F6 (Sklyanin bracket)	proved	cited	specialized	STS 1983
F7 (Gaudin generator)	proved	cited	specialized	FFR 1994

8.2 Loose ends

Three loose ends remain in the seven-face programme:

(i) *Non-algebraic-family constructions*: the master theorem is proved for algebraic families with rational OPE coefficients. The residual universality conjecture restricts to non-algebraic-family constructions, and no known modular Koszul example with simple Lie symmetry falls outside the algebraic-family class.

(ii) *Multi-weight genus $g \geq 2$*: for multi-weight algebras at $g \geq 2$, the face identifications receive a cross-channel correction $\delta F_g^{\mathrm{cross}} \neq 0$. The seven-face programme is the uniform-weight-lane projection; the full multi-weight generalization requires an additional summand from cross-channel stable graphs and is the subject of the multi-weight genus expansion theorem.

(iii) *Non-principal W-algebras*: arbitrary-nilpotent BV duality (bar-cobar commutes with non-principal DS reduction) is conjectural beyond hook-type in type A. The seven-face programme is proved for principal DS reductions and for the hook-type corridor in type A.

The loose ends do not affect the master theorem on the standard landscape; they precisely record the scope boundaries.

8.3 Anti-patterns to avoid

Three anti-patterns expose the common failure modes in using the seven-face programme:

- Never conflate the three invariants $p_{\max}$, $k_{\max}$, $r_{\max}$. Writing "shadow depth" when one means "OPE pole order" is wrong in general: the $\beta\gamma$ system has shadow depth 4 and OPE pole order 1.
- Distinguish generator OPE pole order from composite OPE pole order. The Sugawara tensor of $\beta\gamma$ has quartic self-OPE, but the generators β and γ have simple-pole OPE. The bar complex is built from generators, not composites.
- Do not equate shadow depth with operator order without the $k_{\max}$ translation. The GZ26 operator order is $\max(k_{\max} - 1, 0)$, not $r_{\max} - 2$. The two agree for classes G, L, M but disagree for class C.

9 Closing remarks

The seven-face programme is the unifying thread of the three-volume monograph on chiral algebras and modular Koszul duality. Its content is not a new theorem in any single framework: each face F$i \Leftrightarrow$ Fj identification uses established tools in its source framework. Its content is the *network* of identifications: the assertion that seven different mathematical traditions, working in seven different categorical languages with seven different computational apparatuses, all describe the same underlying element $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]]$ of the Koszul dual.

The surprising aspect is not that equivalences exist: algebraic structures often have multiple presentations. The surprising aspect is that the equivalences close the chain without obstruction, so that starting from any one face and walking the chain returns to the same underlying object. This is the *coherence* of the programme: every face is verified against every other face, both in the mathematics of the monograph and in the 929-test compute layer. Coherence at this scale is not automatic; it is earned, face by face, by checking that the normalization conversions (classical $\hbar \leftrightarrow$ quantum $\hbar$, PVA λ-brackets $\leftrightarrow$ OPE modes, Yangian level shift $\leftrightarrow$ affine level shift) all agree under cross-composition.

The four-class classification G, L, C, M gives the seven faces a structural skeleton. Classes G and L are the "easy" cases: the shadow metric on the primary line is a perfect square, the shadow obstruction tower terminates at degree 3, and the GZ26 Hamiltonians are multiplication operators. Class M is the "generic" case: the shadow obstruction tower is infinite, the Hamiltonians are differential operators of order $2N - 2$, and the seven faces all carry nontrivial information. Class C is the "exceptional" case: the shadow obstruction tower terminates at degree 4 by stratum separation, but the collision residue vanishes, so the GZ26 Hamiltonians are trivial. The $\beta\gamma$ system is the archetype of class C, and it is also the critical witness for the independence of the three invariants.

We close with the observation that the seven-face programme is an *audit instrument*: every time a new face (an eighth framework, a new quantization scheme, a new categorical presentation) is added to the network, its identification with the other seven must be proved separately, and the compute engines must be extended to verify it. The programme is not closed: it is open-ended, and each new face is a new testable prediction. The current statement of Theorem 3.1 is the snapshot at seven; future work will extend it to eight or more.

References

[1] T. Dimofte, N. Niu, A. Py, *Line operators in 3d holomorphic-topological gauge theory and categorified Koszul duality*, preprint 2025.

[2] Z. Khan, P. Zeng, *Poisson vertex algebras and 3d sigma models with holomorphic-topological boundary*, preprint 2025.

[3] D. Gaiotto, P. Zeng, *Holographic reconstruction of chiral algebras from commuting sphere Hamiltonians*, preprint 2026.

[4] V. G. Drinfeld, *Hopf algebras and the quantum Yang-Baxter equation*, Dokl. Akad. Nauk SSSR **283** (1985), 1060–1064.

[5] M. A. Semenov-Tian-Shansky, *What is a classical r-matrix?*, Funct. Anal. Appl. **17** (1983), 259–272.

[6] B. Feigin, E. Frenkel, N. Reshetikhin, *Gaudin model, Bethe ansatz and critical level*, Comm. Math. Phys. **166** (1994), 27–62.

[7] R. Lorgat, *Modular Koszul duality: a monograph on bar-cobar duality for chiral algebras on algebraic curves*, Volume I (manuscript, 2026).

[8] R. Lorgat, *A-infinity chiral algebras and 3d holomorphic-topological QFT*, Volume II (manuscript, 2026).

[9] R. Lorgat, *Calabi–Yau quantum groups and the categorified seven-face programme*, Volume III (manuscript, 2026).